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77-68 PIO-A BOARD

DESIGN NOTE 49

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1. INTRODUCTION

This is a general purpose Parallel Input-Output board for use with the 77-68 system. It can mount one or two 6821 PIA's, and a 6840 Programmable Timer Module, and contains all the necessary system bus buffers and address decoding.

A large 'prototyping' area is provided on the board for the user's particular interfacing circuits, and two 'Scotchflex' 40 way connectors may be mounted on the front edge of the board to carry external connections which the user does not wish to route via the spare contacts of the board 78 way edge connector.

2. BOARD CHARACTERISTICS

Construction ; 8.0" x 3.0" single sided PCB with gold plated edge connector contacts.

Bus ; Buffered, 77-68 CPU compatible.

Power ; Requires +5V DC stabilised @ 0.25A typical

I/O Lines ; 32 parallel data lines, programmable as inputs or outputs.
 8 parallel I/O control lines.
 3 timer clock inputs
 3 timer gate inputs
 3 timer outputs

Addressing ; On board switches select one of 16 address 'blocks' within range F608 - F7FF, allowing multiple PIO-A boards to be fitted in a system. Compatible with 77-68 MON 1 address allocation.

3. CIRCUIT DESCRIPTION

Address Decoding (see Fig 5)

X4 and X5 are both 'Open Collector Output' gates; i.e. each gate output stage is a single transistor with its collector connected to the output pin and its emitter to 0V. As all outputs of X4 and two of X5 are connected to a common line (CS) which is also connected to the +5V line through a resistor (R11), then any one of the gate output transistors turning on will pull CS low. CS will only go high (1) if all of the connected output transistors are in the 'off' state.

When Address Bus lines A3,9,10 & 12-15 are all high (1), X3 output pin 8 will go low, turning off the output transistor feeding X5 pin 10. Similarly, when A11 is low (0), the output transistor connected to X5 pin 12 will turn off.

The output transistor in each of the Exclusive Or gates of X4 will turn off if, and only if, the signals at the gate inputs are different i.e. one '1' and one '0'.

Thus CS will go to '1' when;

- A3,9,10,12,13,14,15 are all '1'
- and - A11 is '0'
- and - A4-7 correspond to the address range selected by switches Sa - Sd.

CS provides one of the Chip Select inputs to X7-9; the others are derived from address lines A2 & A8.

A0,1 (and - for X9 - A2) are also fed to the X7-9 Register Select inputs to determine which of the IC internal registers is to be accessed.

Data Buffers

X6 pin 12 produces a buffered R/W signal which is fed to X7-9 and also enables the data buffers feeding from the system bus to X7-9 when R/W = '0' (CPU Write condition).

When one of the addresses relevant to the board is selected by the CPU, CS goes to '1' as previously mentioned. Then, when the system E line is high, X6 pin 6 goes to '0', enabling those buffers which feed data from X7-9 onto the system data bus.

Interrupt Request

Two of X5 open-collector inverters are used to buffer the IRQ outputs of X7-9. Since low power Schottky logic is used, with a relatively low drive capability compared to standard TTL, the pull-up resistor R14 on the CPU board must be changed to 1k. This will not affect the operation of any other boards on the system.

Refer to Figs 2,3

The constructor should take great care to make good soldered joints using the minimum amount of solder while ensuring that a joint is made all round the component pin or wire end. The use of a small soldering iron with a clean bit is essential.

First examine the track side of the board for any 'swarf' left over from the board drilling process, for unetched copper 'bridges' between tracks, and for possible small breaks in the tracks.

Fit all straps using sleeved wire where appropriate, then form and fit the ribbon cable linking holes e - 1 as shown in Fig 2.

Fit all the capacitors, then the capacitors, and last the IC's (or IC sockets, as preferred).

Finally, clean the track side of the board thoroughly, and examine it for potential short circuits caused by bent leads, excess solder on joints, solder splashes etc.

Change R14 on the CPU board to 1k.

5. TESTING

The following tests are to be done before any custom circuitry is added to the board, and assume the board is set up to respond to addresses F60X (PTM) & F70X (PIA). Note that they do not test all of the PIA and PTM functions, but merely the basic functioning of the board.

a) Check the wiring to the PIO-A board socket, then plug the board into the system and switch on while monitoring the +5V line. Leave switched on for a few minutes while checking all components for signs of overheating (X1,2,7,8 & 9 will get warm, but not hot to the touch).

b) Check that the rest of the system still functions as before, and has not been affected by the addition of the PIO-A board.

c) With a 6321 fitted in the X7 position, use the system monitor to load data into the PAI and read from it in the following sequence;

Read from F709 (Control Reg A), should have been set to 00 by the system reset on turning on.

Write FF to F708 (set Data Direction Reg A for all outputs).

Write 04 to F709 (set CRA bit 2 to '1')

Read from F708 (Peripheral Reg A); should be 00

Write any byte into F708, and check that it can be read back.

d) Connect a 5V DC meter to one of the PIAa outputs PA0-PA7 (pin 2 - pin 9) and check that writing a '1' into the corresponding bit of location F708 gives a 'high' reading (> 2.4V) while writing a '0' gives a reading of less than 0.4V.

e) Repeat tests c & d for the other half of X7 and for both halves of X8, using the appropriate addresses from Fig 5.

f) Connect X9 pin 2 (Counter 2 gate input) to 0V, then:

Write 83 to F609 (sets Control Reg 2 to select E clock, 16 bit continuous-count, interrupt masked, output enabled, and writing to Control Reg 1 enabled).

Write 00 to F608 (clear counter preset)

Write FF to F60C and FF to F60D (Counter 2 latches)

Repeatedly read from F60C (Counter 2 MSB). Differing values should be read as the counter cycles.

Set the CPU running the following program;

20 FE LOOP: BRA LOOP

This generates E pulses for 2 out of every 4 CPU clock cycles.

Monitor the output of counter 2 (X9 pin 3) with a 'scope; it should be a square wave with a half cycle time of approx 200mS (for a 1.6uS cycle time CPU board). If a 'scope is not available then the pulsing waveform can be seen with a moving coil 5V DC meter.

74LS244 (ALC 81LS97)

Bus (TRISTATE) DRIVER

Different Pin-out.

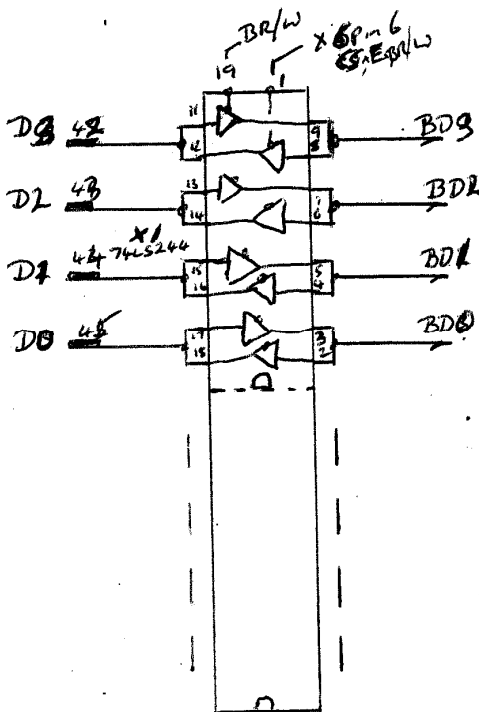
Removed cross links (2-17, 3-18 etc)
joined 2-3, 4-5, 6-7, 8-9

11-12, 13-14, 15-16, 17-18

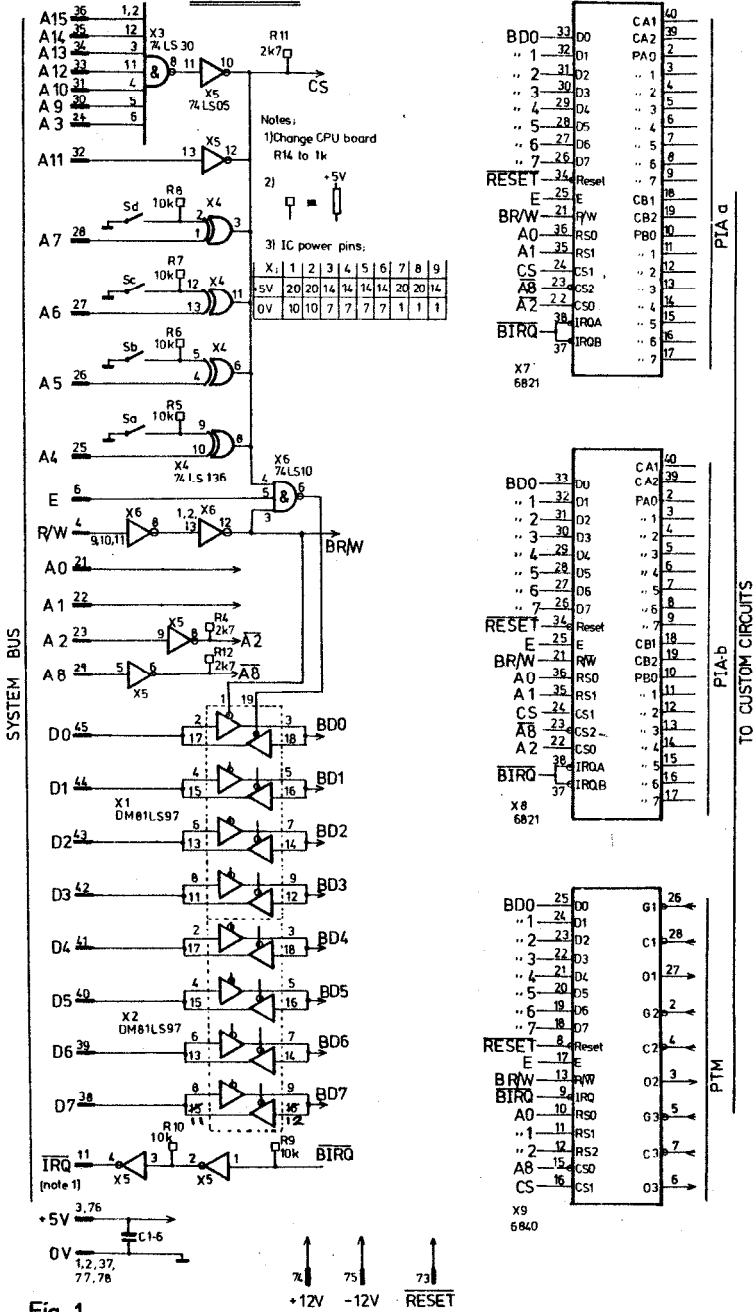
swapped signals from 1 & 19

VCC remains on 20

0V remains on 10



77-68 PIO-A



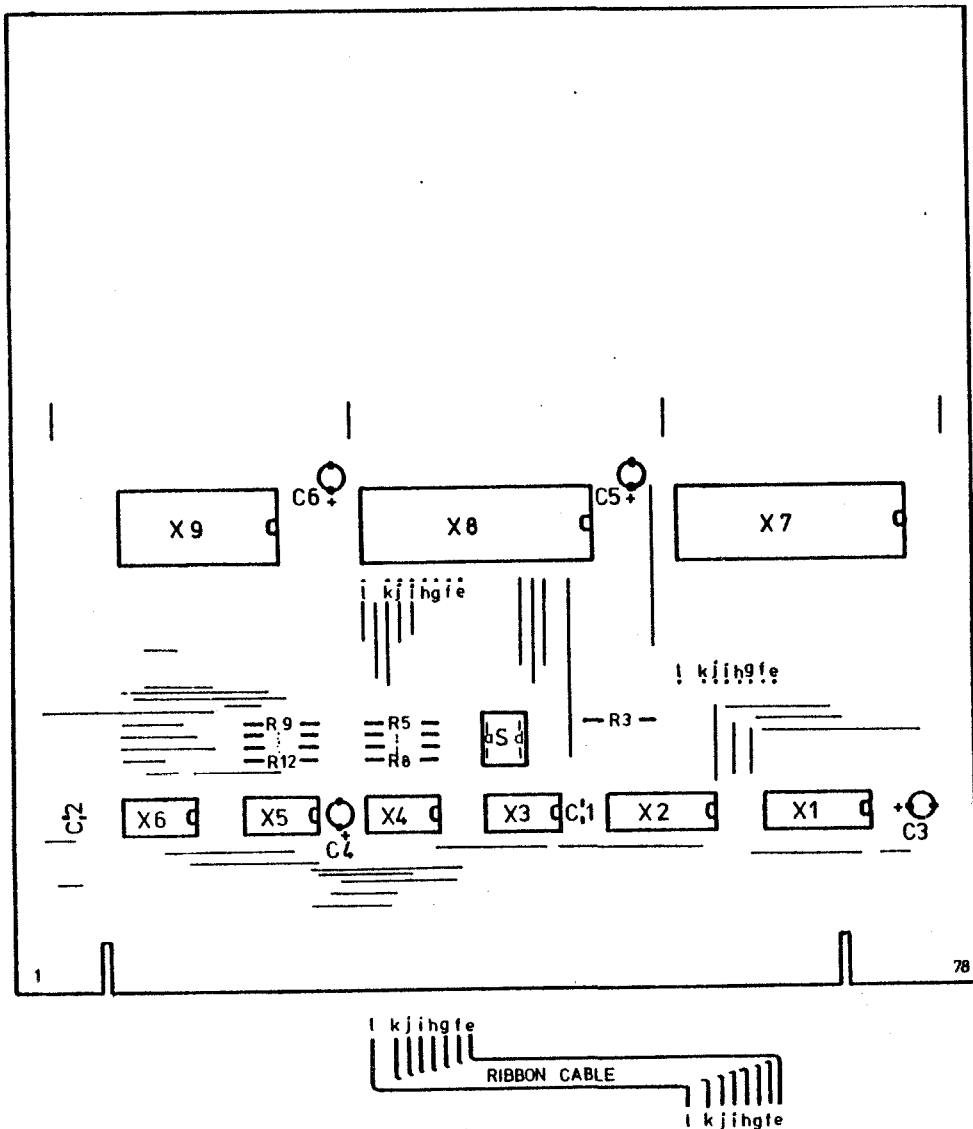


Fig 2 PIO-A BOARD

77-68 PIO-A BOARD

Components

IC's

X1,2	DM81LS97	octal tri-state buffers
X3	74LS30	eight i/p nand
X4	74LS136	quad 2 i/p exclusive or, o/c o/p
X5	74LS05	hex o/c inverter
X6	74LS10	triple 3 i/p nand
X7,8	6821	PIA
X9	6840	Programmable Timer

Resistors

R1,2,3	Not used	
R4,11,12	2k7 ohm	"
R5-10	10k ohm	"
1 off	1k ohm	(to replace existing CPU board R14)

Capacitors

C1,2	0u1 ceramic
C3-6	4u7 16V tantalum bead

Misc

S1 ; 4 pole single throw on-off DIL switch

DIL sockets; if desired; 2 off 40 pin
1 off 28 pin
2 off 20 pin
4 off 14 pin

Ribbon cable - miniature solid cored type

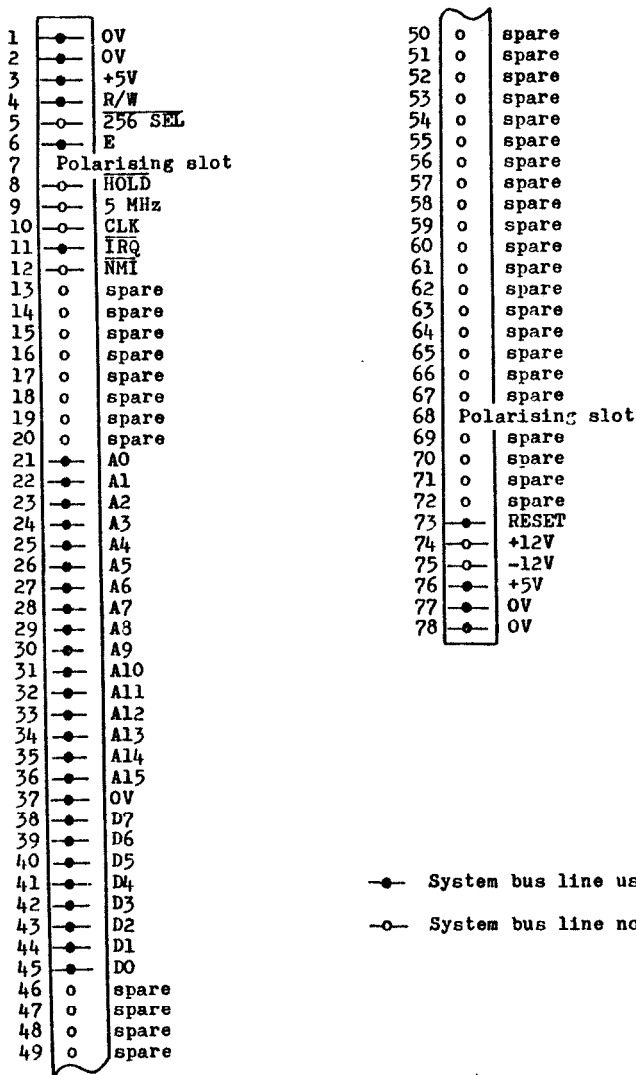
77-68 PIO-A printed circuit board

Mating edge connector socket; 0.1" single sided 78 way (76 connections plus polarising keys)

Note

X7,8,9 will be fitted as required

Fig 3 Component List



EDGE CONNECTOR WIRING

Fig 4

BIT MAP

ADDRESS LINE:	A	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PIAa (X7)		1	1	1	1	0	1	1	1	n	n	n	n	1	0	R	R
PIAb (X8)		1	1	1	1	0	1	1	1	n	n	n	n	1	1	R	R
PTM (X9)		1	1	1	1	0	1	1	0	n	n	n	n	1	R	R	R

n; depends on setting of Sa - Sd

R; selects register in IC

HEX ADDRESS MAP

77-68 system address	PTM (X9) Register selected	
	CPU WRITE	CPU READ
F6n8	Ctrl Reg 3 (CR2 bit 0=0)	-
"	Ctrl Reg 1 (CR2 bit 0=1)	-
F6n9	Control Register 2	Status Register
F6nA	MSB Buffer Register	Timer 1 counter
F6nB	Timer 1 Latches	LSB Buffer Register
F6nC	MSB Buffer Register	Timer 2 counter
F6nD	Timer 2 Latches	LSB Buffer Register
F6nE	MSB Buffer Register	Timer 3 counter
F6nF	Timer 3 Latches	LSB Buffer Register

77-68 system address		Register selected
PIAa (X7)	PIAb (X8)	
F7n8	F7nC	Peripheral Reg A (CRA bit 2 = 1)
"	"	Data Direction Reg A (CRA bit 2 = 0)
F7n9	F7nD	Control Register A
F7nA	F7nE	Peripheral Reg B (CRB bit 2 = 1)
"	"	Data Direction Reg B (CRB bit 2 = 0)
F7nB	F7nF	Control Reg B

'n' depends on setting of switches Sa - Sd;

Sd	Sc	Sb	Sa	n	Sd	Sc	Sb	Sa	n
0	0	0	0	0	1	0	0	0	8
0	0	0	1	1	1	0	0	1	9
0	0	1	0	2	1	0	1	0	A
0	0	1	1	3	1	0	1	1	B
0	1	0	0	4	1	1	0	0	C
0	1	0	1	5	1	1	0	1	D
0	1	1	0	6	1	1	1	0	E
0	1	1	1	7	1	1	1	1	F

'1' = closed, '0' = open

77-68 PIO-A BOARD ADDRESSES

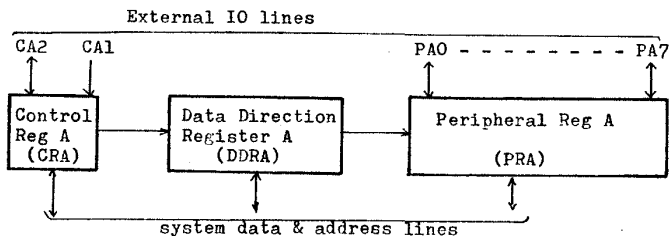
Fig 5

Appendix 1 6821 PIA USE

This appendix gives a brief description of the 6821 operation. See the 6821 data sheet for a fuller description.

Each 6821 contains two almost identical sets of circuits; 'side A' and 'side B'. The 'A' side is described first, then the differences between the A and B sides noted.

Block Diagram of A side of PIA



The A side of the 6821 contains three 8 bit registers:

- Control Register A (CRA)
- Data Direction Register A (DDRA)
- Peripheral Register A (PRA)

The Data Direction Register controls the use of the PA0-7 lines as inputs or outputs. Writing a '1' to a bit of the DDRA causes the corresponding PA line to act as an output (capable of driving 2 standard TTL loads), while writing a '0' causes it to act as an input.

A CPU Read from PRA reads the logic state on the 8 PA lines. ('1' = high) regardless of the state of the Data Direction Register.

Writing to the PRA sets the state of those PA lines which have been programmed as outputs. In fact the data is latched internally, so that altering the DDRA contents can immediately cause data to be output on lines which had previously been programmed as inputs.

The Control Register (CRA) bits are used as follows;

7	6	5	4	3	2	1	0
IRQA1	IRQA2	CA2 control			DDRA access	CA1 control	

read only interrupt flags

read/write

CRA bit 7 (IRQA1) is set to '1' on;

- falling edge (1 to 0) on CA1 if CRA bit 1 = 0
- rising " 0 " 1 " " " " " = 1

In either case, CRA bit 7 going to '1' will cause an interrupt (IRQ) if CRA bit 0 is set to '1'.

CRA bit 6 (IRQA2) is set to '1' when CRA bit 5 is '0', and there is a;

- falling (1 to 0) edge on CA2 in if CRA bit 4 = 0
 - " 0 " 1 " " " " " " " " = 1

In either case, CRA bit 6 going to '1' will cause an interrupt (IRQ) if CRA bit 3 is '1'.

CRA bit 5 set to '1' causes the CA2 line to act as an output with the following characteristics;

CRA bit 4	CRA bit 3	CA2	
		Cleared low	Set high
0	0	After MPU Read from PRA	When CRA bit 7 set to 1 by CA1 input transition.
0	1	After Read	At end of next E pulse which does not address PIA
1	0	As long as CRA bit 3 = 0	-
1	1	-	As long as CRA bit 3 = 1

CRA bit 2 selects access to the Peripheral Register A or the Data Direction Register ; see Fig 5.

B SIDE

Is controlled similarly except for use of the CB2 line as an output, in which case (CRB bit 5 = 1);

CRB bit 4	CRB bit 3	CB2	
		Cleared low	Set high
0	0	After MPU Write to PRB	When CRB bit 7 set by CB1 input transition
0	1	After MPU Write to PRB	After E pulse which does not address PIA
1	0	As long as CRB bit 3 is 0	-
1	1	-	As long as CRB bit 3 is 1

RESET input sets all bits in all registers to zero.

Appendix 2 6840 PTM USE

This appendix gives a brief description of the 6840 Programmable Timer Module, but it is recommended that the user obtains a copy of the 6840 data sheet for detailed reference.

The 6840 contains;

- Three 16 bit binary counters.
- Three corresponding control registers.
- A single Status register.

Associated with each counter are two latches; MSB (Most Significant Byte) and LSB. These hold the data for setting the counters.

A MSB Buffer register is provided also as a temporary store for data being written into the MSB Latches. (Although it can be accessed by three addresses, there is in fact only one MSB Buffer Register). This allows the MSB and LSB Latches to be loaded simultaneously with two bytes by using the following procedure;

- a) Write high byte to MSB Buffer Reg.
- b) Write low byte to Counter Latch (LSB). This causes the contents of the MSB Buffer Reg to be transferred to the corresponding MSB Latch at the same time.

Once loaded, the counter will count down towards zero, at which point it will either stop or (if the continuous mode has been selected) be loaded again from the latches. In continuous mode the output pin will change state as the counter reaches zero, and in any mode an interrupt may be generated if the flag has been enabled.

Similarly, a LSB Buffer Register is used as a temporary store for the low byte when reading from the counters. Thus to read the 16 bit value of a counter;

- a) Read the MSB (high byte) from the appropriate counter. This will automatically place the corresponding low byte into the LSB Buffer Register.
- b) Read the low byte value from the LSB Buffer Register.

Note that writing and reading 16 bit values to and from a counter can therefore be done by the STX,LDX,STS & LDS instructions.

Control Registers

Bit 0 Has different effects according to the control reg;

- CR1; Bit 0 = '1' holds all timers in preset state.
- CR2; Bit 0 controls selection of CR1 or CR3, see Fig 5.
- CR3; Bit 0 = '1' causes Counter 3 clock input to be prescaled (divided) by 8.

The other bits of the three control registers have the same effect for each register;

Bit 1 If '0' then Counter uses CX input as clock.
If '1' then Counter uses E input as clock. This mode is not recommended on the 77-68 as the system 'E' signal rate varies with the type of instruction being executed by the CPU.

Note also that the system CLK signal can only be used by the PTM if it is fed via the divide by 8 prescaler to Counter 3.

Bit 2 If '0' Counter is in normal 16 bit mode.
If '1' then Counter operates in special dual 8 bit mode.

Bits 3,4,5

These define the Counter operating mode. The chief mode being when Bits 3,4 & 5 are all '0', and the counter operates continuously, giving a square wave output with a half cycle = $(N + 1)T$ where N is the 16 bit binary number in the counter latches, and T is the clock period. Single shot, frequency and pulse width comparison modes are also possible ; refer to the 6840 data sheet.

Bit 6 If '1' then a system Interrupt Request is generated when the corresponding Interrupt Flag bit of the Status Reg goes to '1'.

Bit 7 If '1' enables the IC Counter output pin.

Status Register

Bits 0,1 & 2

Are assigned to Counters 1,2 & 3 respectively as Interrupt Flag bits (set when the counter reaches zero and cleared by;

Read Status Reg followed by a Read Counter.
or Write Counter.
or Reset.)

Bits 3,4,5,6

Are not used, and are read as zeroes.

Bit 7 is set to '1' if any of the counters had caused a system interrupt request. It is cleared by;

Read Status Reg followed by a Read Timer.
or Reset.

Reset Input

A low on the system reset line causes;

- All counter latches to be set to their maximum values.
- All Control Reg bit set to '0', except CR1 bit 0 (internal reset bit) which is set to '1'.
- All counters preset to the contents of their latches.
- All counter outputs reset and clocks disabled.
- All Status Reg bits cleared.

Internal Reset (CR1 bit 0)

A '1' causes;

- All counters preset to the contents of their latches.
- All counter clocks disabled.
- Counter outputs and interrupt flags reset.

Appendix 3 Interrupt Handling

Although one may arrange for the CPU to continually monitor the status of the various IO devices on a system, it is often more convenient to have the IO chips call attention to themselves by interrupting the CPU when they require servicing, but otherwise let the CPU perform other tasks.

The 6800 CPU has a single Interrupt Request input (IRQ) line, which when pulled low can interrupt the processor and cause it to jump to a special interrupt handler routine provided by the user. The starting address for this routine being loaded into locations FFF8 & FFF9.

As only a single interrupt line is provided, the interrupt handler must decide which of the possible IO devices is actually causing the interrupt, and it will do this by examining their interrupt flag status bits.

Thus, assuming a fully equipped PIO-A board, set for addresses F60x & F70x, then a suitable interrupt handler would be on the lines of;

```
INTHND: LDA A F609    is it PTM interrupting ?
        BNE PTMINT    if so then go to PTM int routine
        LDA A F709    is it PIA a side A IRQA1 ?
        BMI INTAAL    if so to PIAa IRQA1 routine
        ASL A         is it PIA a side A IRQA2 ?
        BMI INTAA2
        LDA A F70B    is it PIAa side B IRQB1 ?
        BMI INTAB1
        ASL A
        etc.
```

Each individual interrupt routine will of course end with a RTI instruction to return control to the main system program. Also, note that in the case of simultaneous interrupts, this routine has a built in priority; e.g. it services the PTM before the PIA's.